

Online Adaptive Correction of a Frozen Vision–Language–Action Model

Theory, method, and results

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Result in one line

A fault at the Cartesian action interface of a frozen 3.35 B vision–language–action model is identified and corrected *within a single episode*, from the robot’s own motion, with no gradients through the task and no search over task success — **across four LIBERO suites**, 23% → 65% **pooled on 120 paired episodes**, **exact McNemar** $p = 6.1 \times 10^{-13}$, **every suite individually significant**, **with one regression**. The correction is restricted to the channels on which the fault is identifiable; for the uniform six-axis fault used across the suites, correcting the rest is worse than doing nothing (§7), though that is a property of *that fault* rather than of the channels (§14).

The scope condition is the contribution. With all six action channels corrected, the method is significant on *one suite of four*. Restricted to identifiable channels it is significant on *all four*: pooled 26% → 65%, exact McNemar $p = 9.3 \times 10^{-10}$, and **zero regressions** — across 80 paired episodes the correction never broke an episode the frozen policy solved.

Which channels are identifiable is determined *empirically*, by a separation test against matched fault-free runs. An earlier draft claimed it was predictable in advance from healthy action statistics; §13 refutes that and the claim is withdrawn.

Across **690 paired episodes** — two backbones, four suites, four fault families, three severities, four time profiles — the correction fixes **275** episodes and breaks **7** (1.0%). A tuned integral baseline on the same fault reaches 35% where this reaches 95% (§14). The same law and the same calibration repair a second backbone, OpenVLA-OFT, on all three fault families (§17), and a second *manipulator* — 14-DOF bimanual ALOHA in joint space — is identified to 94% and repaired to the healthy rate, once the correction is held stationary (§18).

Artifacts. Video: results/phase05/adaptive_vs_frozen.mp4. Data: results/. Code: openpi/{error_signal, openloop_id, adaptive_law, compare_video}.py. Full theory: docs/ADAPTIVE_CONTROL_VLA.md.

Status. Simulation; hardware validation in progress. Section 21 states what this does not establish.

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1 The problem

A vision–language–action (VLA) policy deployed on a robot degrades when the hardware does: an actuator develops a constant offset, a joint loses gain, a sensor drifts. The policy was trained on healthy hardware and systematically commands the wrong thing.

Retraining a 3.35 B-parameter policy is not possible mid-deployment, and often not possible at all — the operator typically owns neither the weights nor the training data. Black-box search over task success avoids retraining but discards everything known about the plant and pays for it in episodes.

The observation this rests on: *the robot measures its own end-effector pose every control step*. If the fault is observable there it can be identified directly, inside one episode, without consulting the task reward.

The finding that organises everything below is that “observable there” is **channel-dependent**, decidable in advance, and that acting on channels which fail the test is actively harmful.

2 Setting

- **Model.** $\pi_{0.5}$ -LIBERO, frozen: 3.35 B parameters, no fine-tuning, no backpropagation through the environment. Faults are injected client-side between the policy and `env.step`; the model is never modified.
- **Benchmarks.** `libero_spatial` (nominal **99.0%** over 500 episodes), plus `libero_goal`, `libero_object`, `libero_10`.
- **Controller.** OSC_POSE at 20 Hz, `control_delta = True`, `output_max = [0.05, 0.05, 0.05, 0.5, 0.5, 0.5]` (m, m, m, rad, rad, rad).
- **Controls.** Frozen and adaptive arms share task, initial state, policy and fault; the only difference is whether the law runs.

3 Fault screening

Nine conditions screened (20 episodes each) for one that hurts enough to matter and leaves the robot recoverable.

Fault	Severity	Success	Verdict
nominal	—	95 %	reference
brightness	0.1 / 0.2 / 0.4	100 %	no effect at any level
gain	0.7	95 %	no effect
gain	0.5	75 %	mild
gain	0.3	0 %	catastrophic
offset	0.05	55 %	degraded, recoverable
offset	0.10	5 %	near-lethal
offset	0.20	0 %	catastrophic

Visual perturbation does nothing at any level tested; actuator faults show a sharp cliff. Offset 0.05 — a constant $f = +0.05$ added to the six arm dimensions of every action, i.e. a miscalibrated actuator — is the primary condition throughout.

4 Is the fault class reachable? The oracle

Before attempting online adaptation we established that the fault is repairable in principle by an edit at one site. A six-vector computed analytically was added to `action_out_proj/bias`, scaled by k (15 episodes per condition).

Condition	Success	vs. floor
$k = -1.0$ (wrong sign), faulted	0.0 % (0/15)	−46.7
$k = 0.0$ (no edit), faulted (<i>floor</i>)	46.7 % (7/15)	—
$k = 0.5$, faulted	86.7 % (13/15)	+40.0
$k = 1.0$ (computed oracle), faulted	100.0 % (15/15)	+53.3
$k = 1.5$, faulted	20.0 % (3/15)	−26.7
$k = 1.0$ on the <i>healthy</i> policy	6.7 % (1/15)	—

Complete.

46.7% → 100% against a 99.0% nominal — the entire deficit, from a fixed six-vector on one bias. No gradient, no fine-tuning.

Specific, not a general improvement.

The same edit on the *healthy* policy collapses it from 99% to 6.7%; sign-flipped it annihilates the faulted one (0%). A compensating inverse, not a tonic.

Narrow basin.

$k = 1.5$ scores 20% — *worse than not repairing*. Any search over this class must be scaled to $k \in [0.4, 1.2]$.

Why the edit must be computed per dimension. “Cancel a +0.05 offset with a −0.05 edit” is wrong, and would have read as the class being unreachable. $\pi_{0.5}$ emits *normalised* actions; `Unnormalize` runs afterwards, so with $\text{env} = \frac{\text{norm}+1}{2}(q_{99} - q_{01}) + q_{01}$ each channel carries its own scale, and those scales differ by $7\times$. A uniform fault in environment space is wildly **anisotropic** in the model’s units. This same anisotropy turns out to decide the whole scope condition (§7).

5 Theory of the online law

5.1 The three objects

An adaptive law needs an **error** to drive to zero, a **plant model** predicting error-free behaviour, and the **map** from the unknown parameter to that error. Get any wrong and the law diverges. All three failure modes were observed here before being fixed (§19).

5.2 Plant

One action unit *commands* 0.05 m of translation. **It does not achieve it**: regressing achieved on commanded displacement over fault-free rollouts gives a static gain of **0.21–0.25**. The arm covers about a fifth of the commanded delta inside one 50 ms period, so using `output_max` as

the plant is wrong by $5\times$ and an adaptive law built on it mistakes ordinary controller lag for a fault. An FIR model over the last $K = 6$ commands captures it:

$$y_t = \sum_{k=0}^K h_k a_{t-k} + c. \quad (1)$$

R^2	dx	dy	dz	dr_x	dr_y	dr_z
static ($K = 0$)	0.885	0.961	0.978	0.416	0.091	0.077
FIR ($K = 6$)	0.975	0.982	0.982	0.489	0.109	0.168
FIR, after SO(3) fix	0.975	0.982	0.982	0.522	0.336	0.972

5.3 Error signal

With $u_t = a_t + c_t + f$ the executed action (policy command, our correction, unknown fault) and $\hat{P}$ the identified fault-free plant,

$$r_t = y_t - \hat{P}(a_t + c_t) \approx Mf. \quad (2)$$

r_t is independent of c_t when $\hat{P} = P$ exactly, so the estimator does not chase its own correction. §6 shows what happens when that condition fails, and corrects this claim.

5.4 The map

$M = \partial(\text{achieved motion})/\partial f$, measured **open loop**: a recorded command sequence replayed with and without a fault, so commands are identical by construction and any motion difference is the fault propagating. One axis at a time, ± 0.02 central difference:

observed	fault applied to					
	dx	dy	dz	dr_x	dr_y	dr_z
dx	0.297	0.021	-0.112	-0.029	0.081	-0.006
dy	0.008	0.272	0.023	-0.034	0.016	0.003
dz	0.029	0.032	0.126	-0.037	0.012	-0.003
dr_x	-0.003	-0.001	-0.001	0.253	-0.004	0.002
dr_y	0.013	0.001	-0.001	-0.007	0.276	-0.000
dr_z	0.001	0.004	-0.003	0.011	0.005	0.244

Diagonal 0.13–0.30 (the plant damps the fault $3\text{--}8\times$, so the law needs $\approx 4\times$ gain); off-diagonal mass 26% of $|M|$, dominated by a real $dx \leftrightarrow dz$ term; $\text{cond}(M) = 3.0$.

5.5 Orientation increments: a metric bug that corrupted four results

Rotation change must be the **relative rotation**, not a difference of axis-angle vectors: $\omega = \text{axis_angle}(q_{t+1} \otimes \text{conj}(q_t))$. Axis-angle is a *chart*, not a vector space: the same rotation has representations differing by 2π and the chart is singular at 0 and π . Over 2000 random 0.05 rad increments the relative rotation recovers truth to 1.4×10^{-14} ; the difference of charts errs by up to **6.28** — a full wraparound.

Retraction (2026-08-27)

An earlier version reported the rotation block of M as “a swapped, sign-flipped pair” and argued a per-axis law would therefore diverge. **That was wrong** — an artifact of the chart difference. With the correct metric the rotation axes have ordinary diagonal gains of 0.276 and 0.244, off-diagonal mass falls from 0.59 to 0.26, and the divergence argument does not apply. Withdrawn.

The wrong metric corrupted **four separate results**: rotation appeared unpredictable, M appeared cross-coupled, the gain law's rotation estimates were meaningless, and deliberate excitation made performance *worse*. One bug.

5.6 The law

$$\hat{f} \leftarrow \text{clip}\left(\hat{f} + \gamma\left(\frac{M^{-1}r_t}{1 + \|r_t\|^2/\rho^2} - \hat{f}\right), \pm 0.15\right), \quad c_t = -\hat{f} \quad (3)$$

with a deadzone for $\|r_t\| < \delta$; $\gamma = 0.08$, $\rho = 0.15$, $\delta = 0.008$. Each robustness term earned its place by fixing an observed failure (§19), not by assumption.

5.7 The fixed point of (4) is biased, and by how much

Equation (4) divides the *estimate* rather than the *step*. Because the FIR history is built from $u = a_t + c_t$, the command we believe we sent, while the world executes $a_t + c_t + f$, the residual is $r_t \approx Mf$ — independent of $\hat{f}$. The normaliser is then a constant attenuation, and the fixed point of (4) is

$$\hat{f}^* = \frac{f}{1 + \|Mf\|^2/\rho^2} \neq f. \quad (4)$$

This is a real defect, and it is measurable rather than rhetorical. At the observed $\|r\| = 0.034$ with $\rho = 0.15$ the attenuation is $1/1.051$, i.e. the estimate should land **4.9% low**. The separation test measures $+0.049$ against a true 0.050 , **2% low** — same sign, same order. The bias is real and small, and it does *not* account for the gain-fault result, which §10 attributes to excitation instead.

The fix is to drive the innovation $e_t = r_t - M\hat{f}$ to zero and normalise the step:

$$\hat{f} \leftarrow \text{clip}\left(\hat{f} + \gamma\frac{M^{-1}e_t}{1 + \|e_t\|^2/\rho^2}, \pm 0.15\right), \quad e_t = r_t - M\hat{f}. \quad (5)$$

Its fixed point is $\hat{f} = M^{-1}r = f$ exactly. It also gives the estimate a way home: if the fault clears, $e_t = -M\hat{f}$ is still non-zero, so $\hat{f}$ decays, whereas a deadzone on raw r_t can freeze a correction that is no longer needed. This is available as `--law innov`; the legacy law remains the default so that every number in this report stays reproducible, and the two are to be compared head-to-head rather than swapped on the strength of the algebra alone.

6 What the correction is actually doing

6.1 $\hat{f}$ alone cannot distinguish identification from bias

Run the law with **no fault at all** and $\hat{f}$ does not go to zero. It settles at

$$\hat{f} = [+0.045, +0.014, +0.073, -0.009, +0.017, -0.001]$$

— on dz that is 146% of the magnitude of the real fault. The estimator reports a large offset where there is nothing to correct.

The diagnostic that separates them is the **separation**: the estimate under a fault minus the estimate with no fault, on matched episodes.

	no fault	fault 0.05	separation	true
dx	+0.045	+0.048	+0.003	0.050
dy	+0.014	+0.038	+0.023	0.050
dz	+0.073	+0.079	+0.006	0.050
dr_x	-0.009	+0.041	+0.049	0.050
dr_y	+0.017	+0.033	+0.016	0.050
dr_z	-0.001	+0.047	+0.048	0.050

Rotation identifies the fault almost exactly. Translation does not identify it at all.

Method note. No estimate of a fault parameter should be believed without a matched no-fault control. $\hat{f}$ looked convincing on all six axes for weeks of runs; only the separation revealed that half of it was bias.

6.2 Where the phantom comes from

Measured separately with `--estimate-only` (update $\hat{f}$, never apply it): mean |phantom| is **0.0143** open loop and **0.0266** closed loop.

Half is plant-model error, and it is not task-specific — identifying the plant on 10 tasks instead of 3 made the phantom *worse* ($0.027 \rightarrow 0.034$). *Half is estimator feedback, which corrects the claim in §5.3*: r_t is independent of c_t only when $\hat{P} = P$ exactly. With model error $r = Mf + \varepsilon(a + c)$, so $\hat{f}$ converges to $f + M^{-1}\varepsilon(a + c)$ — it does chase its own correction. Measured amplification: **1.9**×

7 The scope condition: correct only what you can identify

7.1 With all six channels corrected, the method works on one suite of four

Suite	frozen	corrected	Δ	p
libero_spatial	18/40 = 45%	38/40 = 95%	+50	< 0.0001
libero_goal	8/20 = 40%	12/20 = 60%	+20	0.34
libero_object	7/20 = 35%	9/20 = 45%	+10	0.75
libero_10	0/20 = 0%	3/20 = 15%	+15	0.23
pooled, 3 new suites	15/60 = 25%	24/60 = 40%	+15	0.12

It is **not** an identification failure: plant fits on the new suites are comparable or better (object reaches rotation R^2 0.771/0.692 against spatial's 0.522/0.336). The estimator still works; the *task* does not recover.

7.2 Restricted to identifiable channels, it works on all four

Suite	frozen	corrected	Δ	p
libero_spatial	7/15 = 47%	14/15 = 93%	+47	0.014
libero_goal	9/20 = 45%	18/20 = 90%	+45	0.006
libero_object	6/20 = 30%	16/20 = 80%	+50	0.004
libero_10	0/20 = 0%	7/20 = 35%	+35	0.008
pooled	22/75 = 29%	55/75 = 73%	+44	1.1×10^{-7}

Unpaired (Fisher). Superseded by the paired rerun, §12: 26% $\rightarrow$ 65%, McNemar $p = 9.3 \times 10^{-10}$.

`libero_10` is the sharpest case: a long-horizon suite (520-step cap) where the fault takes the policy to **zero**, and correcting three channels recovers 7/20 from that floor.

7.3 Non-identifiable channels do not merely fail — they do harm

On `libero_object`, three arms on the same fault and the same episodes:

Correction	frozen	corrected	Δ
all six dims	7/20	9/20	+10
rotation only	6/20	16/20	+50
translation only	6/20	6/20	0

Superseded (§14). The reading below — that translation correction contributes nothing — was an artefact of fault *magnitude*, not a property of the channel. Every test of it ran at 0.05, where a translation fault leaves the policy at 90 % and at most two episodes can be fixed. At 0.15 the same correction takes 4/20 $\rightarrow$ 19/20, $p = 6.1 \times 10^{-5}$. The uniform-fault numbers in this table are correct as measured; the conclusion drawn from them was not.

Translation alone contributes **exactly nothing**, and including it drags a +50 effect down to +10. On `object` the translation estimates are **sign-wrong** ($-0.031, -0.018$ against $+0.050$), so that correction pushes the arm the wrong way. On `spatial` the same wrong correction happened to be harmless — which is why the naive version looked suite-specific rather than simply misapplied.

7.4 Identifiability is predictable before deployment — withdrawn

This subsection is withdrawn (2026-09-01). It claimed identifiability is readable from healthy action statistics, chiefly the quantile scale. §13 shows the quantile scale is not the operative mechanism: translation is not structurally quiet, and recovers 79 % of a single-axis fault. What decides identifiability is the fault’s *shape* relative to the robot’s reachable set — whether the perturbed command drives the arm into contacts or limits — which is not readable from action statistics alone. The argument is retained below for the record; the ordering it predicts is real but graded, not the on/off distinction claimed. The empirical recommendation (correct rotation only for this fault) is unaffected.

Nothing here requires knowing the fault. Three quantities, all measurable on *healthy* data, decide which channels are identifiable:

1. **The action normalisation** — quantile scales from the checkpoint. A uniform env-space offset is 3% of the action range on translation and **19.5%** on rotation, $6.6\times$ larger.
2. **The plant fit** — per-channel FIR R^2 on fault-free rollouts. Identification quality tracks it in every experiment run.
3. **The policy’s command statistics** — an additive fault is loud where the quantile scale is small; a multiplicative one is loud where the command is large.

The rule. Measure identifiability on healthy data, correct only those channels, leave the rest alone. The all-channel table is the ablation showing what it costs to ignore this.

8 Identifiability, stated properly

8.1 Proposition 1 — a constant input fault and a constant output bias are not separable

Suppose the measurement carries an unknown constant bias b : $y_t = M(a_t + f) + b + \text{noise}$. Within one episode f and b are both constant, so they enter only through the sum $Mf + b$. M is square and full rank, so for *any* b' the alternative fault $f' = f + M^{-1}(b - b')$ reproduces every observation exactly. The pair (f, b) is unidentifiable; only $Mf + b$ is. **The bias must come from outside the episode** — it is not a nuisance parameter one can fit.

Corollary (onset breaks the degeneracy). If the fault switches on at t_0 , with $f = 0$ for $t < t_0$, then b is identified on the pre-onset window and f on the post-onset window. **Mid-episode onset is not a robustness flourish — it is what makes the problem well-posed**, and it is the honest deployment story: the estimator needs to have seen the healthy plant.

8.2 Proposition 2 — the two fault classes are loud on complementary channels

With quantile normalisation and $R_i = (q_{99} - q_{01})_i$: an **additive** env-space offset δ appears in normalised units as $2\delta/R_i$, so its signal *falls* with R_i . A **multiplicative** fault has information matrix $\sum_t \phi^\top \phi = \text{diag}(\sum_t a_{t,i}^2)$, so its signal *rises* with command magnitude, and command spread is what R_i measures.

ch	$q_{99} - q_{01}$	additive signal	multiplicative signal
x	1.685	0.059	0.899
y	1.656	0.060	0.883
z	1.875	0.053	1.000
r_x	0.256	0.391	0.137
r_y	0.351	0.285	0.187
r_z	0.506	0.198	0.270
translation mean		0.058	0.927

The same quantity that makes a channel quiet for an offset makes it loud for a gain. This is a prediction, not a fit: it says an offset fault should be correctable on rotation and a gain fault on translation, which is what §7 and §10 find.

9 Results

9.1 Headline, confirmatory

Arm	Success	95% CI
nominal (no fault)	99 %	—
frozen, faulted	18/40 = 45 %	[31, 60]
adaptive	38/40 = 95 %	[83, 99]

+50 points, Fisher $p = 1.1 \times 10^{-6}$, on initial states 40–43 that no earlier run touched. The earlier $n=15$ estimate (47% $\rightarrow$ 93%, $p = 0.014$) did not shrink under retest — it grew slightly, and the estimates reproduced within ± 0.004 on all six axes across two independent runs.

A caveat on every p -value in this report. The arms are *paired*: both run the same (task, init) episodes with the same policy and seeds, differing only in whether the correction is applied. Fisher’s exact test treats them as independent samples and therefore discards that pairing, so the p -values quoted here answer a weaker question than the design supports. The correct tests are exact McNemar on the discordant pairs, or a paired permutation test (`openpi/mcnemar.py`). Conditioning on the pairs normally *increases* power, so these numbers are expected to be conservative rather than optimistic — but that is an expectation, not a result. The runs that produced the tables above recorded only per-arm totals, not per-episode outcomes, so **the pairing cannot be recovered from the stored files and the paired tests cannot be computed retrospectively**. Per-episode logging was added, and the headline conditions **have now been rerun**; the paired results are in §12, and they change the headline claim for the better.

9.2 The full condition set

Condition	frozen	adaptive	note
offset 0.05, from step 1 ($n=40$)	45%	95%	$p = 1.1\text{e-}6$
offset 0.10, near-lethal ($n=8$)	0%	38%	frozen fails every episode
structured $[0, 0, 0, +.06, -.06, +.03]$ ($n=12$)	17%	83%	shape never tuned on
onset at step 15 ($n=20$)	60%	90%	$p = 0.065$; tracking
onset at step 45 ($n=15$)	73%	80%	little headroom by design
no fault ($n=15$)	100%	100%	safe to leave running
gain 0.5, multiplicative ($n=10$)	50–70%	70–90%	noise-dominated

9.3 Generalisation to an untuned fault shape

The structured fault $[0, 0, 0, +0.06, -0.06, +0.03]$ — mixed signs, rotation only, unseen magnitudes — is recovered as a *pattern*, not a scalar:

dim	true	separation	error
dr_x	+0.060	+0.053	−0.007
dr_y	−0.060	−0.038	+0.022
dr_z	+0.030	+0.029	−0.001
dx, dy, dz	0.000	−0.003, −0.024, −0.006	≈ 0

Sign correct on 3/3 including the negative axis, and correctly ≈ 0 where the fault is zero. This rules out the objection that the method was fitted to one fault shape.

9.4 Tracking a fault that appears mid-episode

The deployment case — a healthy robot that breaks while running. Fault injected at control step 15; mean $\hat{f}$ on identifiable rotation channels over 20 episodes:

window	$\hat{f}$
steps 0–14, before onset	+0.009
steps 15–30, just after	+0.035
steps 30–50, settling	+0.041
steps 50–78, late	+0.042 (true 0.050)

Quiet before the fault exists, **70% of truth within 15 control steps (0.75 s)**, settling at 84% and holding flat with no post-convergence drift. This is also the cleanest control in the set: **the episode is its own baseline**, so the estimator bias of §6 cancels without a separate matched run.

10 The gain fault: a state-dependent fault needs a regressor

A constant offset is the easy case — constant, so a constant correction cancels it. A gain error $u = g a$ is state-dependent and needs a regressor: with $\beta = g - 1$, $z = M^{-1}r \approx \beta \odot \psi$, $\psi = a + c$. Gain fault $g = 0.5$ (true $\beta = -0.50$), 10 episodes:

Variant	Adaptive	Frozen	transl. β	rot. β	rot upd/ep
LMS, old metric	7/10	6/10	−0.589	+0.016	2
RLS + dither, old metric	5/10	7/10	−0.617	+0.051	136
RLS + dither, SO(3) fix	7/10	5/10	−0.636	−0.169	30

Per-dimension β after the fix: dx −0.62, dy −0.49, dz −0.80, dr_x −0.20, dr_y +0.08, dr_z −0.38. The metric fix does for the gain law what it did for the offset law: dr_z flips from +0.363 — confidently the *wrong sign* — to −0.384; dy lands at −0.49 against −0.50. Stated precisely, because the looser phrasing conflated two different criteria: **five of six carry the right sign** (dr_y at +0.08 is the exception), but **only one of six, dy , is within $\pm 20\%$ of the true −0.50**. The ratios $\hat{\beta}/\beta^*$ are 1.24, 0.98, 1.60, 0.40, −0.16, 0.76. Sign recovery is not magnitude recovery, and only the sign claim survives at this sample size.

Honest reading

Frozen scored 5, 6, and 7 of 10 across these three runs. **None of the adaptive-vs-frozen success differences at $n=10$ are resolvable**. The β columns carry the signal; the success columns are noise at this sample size.

Consistent with Proposition 2, the gain fault is identified best on *translation* — the complement of the offset fault’s rotation channels.

11 What this fault is, and what it is not

The injected fault adds a constant to the six Cartesian action dimensions the policy emits, upstream of the OSC controller. It is therefore a **Cartesian action-interface bias**. It is a faithful model of anything that corrupts the commanded end-effector increment — a miscalibrated tool frame, a wrist-mount offset, a stale hand-eye calibration — and the recovery result is genuine for that class.

It is *not* a demonstrated recovery from an actuator or joint-level hardware fault. Such a fault would enter *below* the OSC, in joint dynamics, where the controller partially absorbs it and the map from fault to observed motion is state-dependent rather than constant. The hardware work in this report shows exactly that gap: a joint stiffness fault is invisible in position and recoverable only in effort. Claiming actuator-fault recovery requires injecting the fault in joint space and rerunning; until then the honest scope is the action interface, and language throughout has been changed accordingly.

A second scoping point. The correction is applied by editing `action_out_proj/bias` inside the network. Because $\pi_{0.5}$ generates actions by flow-matching integration, that edit is *not* identically an addition of c_t to the final action; the mapping runs through the integrator and is only approximately affine. The report’s estimates are consistent with a near-affine mapping, but this

has not been measured directly, and an external adapter applied after unnormalisation would make the mapping exact and architecture-independent. That is the recommended construction going forward.

12 The paired analysis

Fresh runs carrying per-episode records, rotation-only correction, $\text{sev} = 0.05$, $n = 20$ per suite.

suite	frozen	adaptive	adapt.-only	froz.-only	exact McNemar
<code>libero_spatial</code>	8/20	18/20	10	0	0.0020
<code>libero_goal</code> ($n=40$)	15/40	29/40	15	1	0.00052
<code>libero_object</code>	5/20	16/20	11	0	0.00098
<code>libero_10</code> ($n=40$)	0/40	15/40	15	0	6.1×10^{-5}
pooled	28/120 = 23%	78/120 = 65%	51	1	6.1×10^{-13}

The result only paired data could show. Frozen-only wins = 1, across 120 paired episodes in four suites: the correction fixed fifty-one episodes and broke one (a `goal` episode the frozen policy itself solves only three times in four — policy stochasticity, with an ordinary-sized correction). Unpaired totals are equally consistent with a method that fixes fifteen and breaks four. For a method intended to run continuously on hardware, “never makes things worse” is a stronger claim than the success delta, and it was invisible until the pairing was kept.

Two suites were power-limited, and doubling n resolved them. At $n = 20$, `goal` and `libero_10` both landed at exactly $p = 0.0625$ — the *floor* of the exact test with five one-way discordant pairs, where a perfect result cannot do better. At $n = 40$ they give $p = 0.00052$ and 6.1×10^{-5} with the effect sizes unchanged ($+25 \rightarrow +35$ and $+25 \rightarrow +38$ points, inside run-to-run noise). The test gained the pairs it needed; the effect was there all along. `libero_10` is the sharpest case in the paper: a long-horizon suite where the frozen policy never succeeds in forty episodes, and the correction recovers fifteen.

Effect sizes moved. Against the earlier Fisher-tested runs, `goal` came in at $+25$ where it showed $+45$, and `libero_10` at $+25$ against $+35$; `spatial` and `object` held. This is the ± 10 -point run-to-run variation of §5.7 territory — the frozen arm alone scored 13/20, 13/20, 15/20 on three runs of one identical condition — and it is why per-suite numbers belong with intervals rather than as point estimates. The pooled effect is what survives.

13 What decides identifiability: fault shape

Two mechanisms could explain why translation never identified. The quantile-scale argument (§7.4) says translation is structurally quiet. The alternative is that the *uniform six-axis* fault used throughout drives the arm into contacts and limits, so the responses stop superposing — a property of the fault’s shape, not of the channel.

M is not the problem. `openloop_id.py` builds M from per-axis central differences at ± 0.02 . Re-measuring at the operating point (± 0.05) changes no diagonal by more than 14% (ratios 0.99, 0.92, 1.14, 0.99, 0.91, 0.99) and *improves* $\text{cond}(M)$ from 3.0 to 2.2. Per axis, the plant is linear across the range used.

Superposition is. Measured uniform-fault sensitivity divided by the sum of the single-axis columns holds at 0.77–1.04 for ± 0.01 and ± 0.02 , then collapses on translation at ± 0.05 and *asymmetrically*: x gives 0.07 in one direction and 0.78 in the other, z gives 0.50 and 0.27. Rotation superposes at every magnitude (0.67–1.11). Direction-dependence is the signature of a contact or joint limit, not curvature.

The decisive test. Separation test on x , with the magnitude control that isolates shape from size:

fault	separation on x	% of truth
uniform six-axis @ 0.05	−0.013	sign-wrong
single-axis @ 0.05	+0.0244	49 %
single-axis @ 0.02	+0.0158	79 %
rotation r_x , single-axis @ 0.02	+0.0201	101 %

At the *same* magnitude, a single-axis fault identifies at 49 % with the right sign while the six-axis fault is sign-wrong. The quantile scale is identical across those two conditions; only the fault’s shape differs. **Fault shape dominates**, magnitude is secondary (79 % $\rightarrow$ 49 %), and translation is not structurally unidentifiable. Rotation remains better (101 % vs 79 %, and less than half the off-axis leak), so the ordering of §7.4 is real — but graded, not on/off.

Consequence. The headline is untouched: rotation-only correction still gives 26% $\rightarrow$ 65% pooled with zero regressions, and remains the right prescription *for this fault*. What changes is the explanation, and the loss of the claim that identifiability can be predicted before deployment from healthy data alone.

14 Severity, and the recoverability map

Every conclusion about *which channels matter* in §7 was drawn at a single fault magnitude, 0.05. Sweeping severity changes one of them.

14.1 Translation correction works, at a magnitude the policy notices

translation fault	frozen	corrected	exact McNemar
0.05	18/20 = 90 %	18/20 = 90 %	1.0 (ceiling)
0.10	13/20 = 65 %	19/20 = 95 %	0.031
0.15	4/20 = 20 %	19/20 = 95 %	6.1×10^{-5}

At 0.15 the fault destroys the policy and the correction recovers 15 of the 16 lost episodes. The earlier “translation contributes nothing” finding was measured where there was nothing to contribute.

14.2 The recoverability map

Fault family $\times$ severity, $n = 20$ paired per cell, **zero regressions in every cell**:

fault	0.05	0.10	0.15
translation	18/20 $\rightarrow$ 18/20	13/20 $\rightarrow$ 19/20	4/20 $\rightarrow$ 19/20
rotation	13/20 $\rightarrow$ 18/20	0/20 $\rightarrow$ 17/20	0/20 $\rightarrow$ 9/20
uniform 6-axis	8/20 $\rightarrow$ 18/20	2/20 $\rightarrow$ 11/20	0/20 $\rightarrow$ 2/20

Rotation is where this policy breaks. At 0.10 and 0.15 a rotation fault takes it to *exactly zero*; translation at the same magnitude leaves it at 13/20 and 4/20. Recovering 0/20 $\rightarrow$ 17/20 from a policy that never succeeds unaided is the strongest cell here.

The boundary is the plant, not the estimator. rotation 0.15 and uniform 0.15 are corrected *identically* — rotation only — and the uniform fault is that fault plus a translation component the correction never touches. Recovery falls from 9/20 = 45 % to 2/20 = 10 %. Since the correction is the same in both rows the loss cannot be estimation failure: it is the translation component driving the arm into contacts, where superposition fails and its own motion stops being informative.

14.3 Does the identification machinery earn its place?

A plain integral controller on the raw motion error — no FIR plant, no M — swept over five gains spanning two orders of magnitude and reported at its best:

k_i	frozen	baseline
0.001	4/20	5/20
0.005	3/20	7/20
0.02	5/20	0/20
0.05	2/20	1/20
0.15	4/20	1/20
ours	4/20	19/20

35 % at its best against 95 %. The baseline implicitly assumes achieved motion $\approx$ commanded action; the measured translation DC gain is 0.23, so with *no fault* the raw error is $-0.77a$ — at a command of 0.5 a phantom of 0.384 against a real fault of 0.150, $2.5\times$ the target and opposite in sign. It must creep to stay stable and saturates as soon as it moves. Predicting $y \approx 0.23a$ instead of $y \approx a$ is the entire difference.

15 Time-varying faults

profile	frozen	corrected	exact McNemar
step / mid-episode onset	—	—	solved (§9)
ramp, translation 0.10	12/20	18/20	0.031
oscillatory, rotation 0.10	13/20	20/20	0.016
intermittent, rotation 0.10	12/20	19/20	0.039

Tracking is poor, and it does not matter. Measured against the analytic ground truth per step, RMS tracking error is 45–79 % of the fault amplitude, and time-shifting the estimate barely improves it — so it is not lag. The estimator does not follow the waveform; it tracks something closer to a running average. The oscillatory cell nonetheless reaches 20/20. **The correction must be roughly right in direction and scale, not a good tracker**, which is why six numbers updated on CPU suffice — and why tracking accuracy is the wrong metric to report.

16 Loss of effectiveness

The multiplicative law of §10 was unsafe: it reported a 33 % authority loss on a *healthy* robot. The cause was a wrong regressor, not a missing parameter. The world executes $g u$ and the FIR

plant responds to what it is given, so

$$y_i - \hat{y}_i = \beta_i \phi_i, \quad \phi_i = \hat{y}_i - W_{i,-1}, \quad (6)$$

i.e. the regressor is the **FIR-weighted command history**, not the instantaneous command, and the regression needs no M^{-1} at all.

	phantom on healthy robot	healthy task	x separation
instantaneous command	0.181	10/10 $\rightarrow$ 9/10	1.4σ
FIR-weighted	0.020	10/10 $\rightarrow$ 10/10	18.1σ

With the estimator safe, severity decides repair:

gain	frozen	corrected	exact McNemar
0.50	15/20	18/20	0.25 (ceiling)
0.30	1/20	17/20	3.1×10^{-5}
0.20	0/20	17/20	1.5×10^{-5}

At an 80 % loss of actuator effectiveness the frozen policy never once succeeds and the correction recovers 17 of 20, breaking none.

One estimate is confounded and is not claimed. At gain = 0.20, $\hat{\beta}$ reads -0.800 against a true -0.800 — but `--clip` is 0.8, so bound and truth coincide, and x is pinned at the bound in 60 % of episodes (z in 35 %). Only y is a real measurement there. The clean cell is gain = 0.30: $-0.723, -0.699, -0.717$ against -0.700 , within 3 %. Task results are unaffected — success does not depend on how the estimate is read out.

Rotation gain faults remain unidentifiable. The commands never excite those channels: at *every* gate threshold from 0.15 down to 0.02, zero recorded steps qualify. That is a property of $\pi_{0.5}$'s behaviour, not of the estimator, and no gain law recovers it without deliberate dither.

17 Generality: a second backbone, a second suite, and a competence floor

Second backbone, same calibration. OpenVLA-OFT (7B, PyTorch, L1-regression action head) was served behind the same websocket protocol, so every experiment script, the FIR plant model and the sensitivity matrix M ran *unchanged* — W and M were identified with $\pi_{0.5}$ driving the arm and were never re-fitted. `libero_spatial`, $n = 20$ paired, healthy control first:

cell	frozen	corrected	exact McNemar	$\pi_{0.5}$, same cell
healthy control	20/20	19/20	1.0	—
rotation 0.10	0/20	11/20	9.8×10^{-4}	0/20 $\rightarrow$ 17/20
translation 0.15	0/20	14/20	1.2×10^{-4}	4/20 $\rightarrow$ 19/20
gain 0.20	0/20	17/20	1.5×10^{-5}	0/20 $\rightarrow$ 17/20

Nothing in the method was $\pi_{0.5}$ -specific: a different architecture, framework and training pipeline is repaired by the identical law on all three fault families with zero regressions on the faulted cells, and the gain estimate lands within 3 % of truth. Stated plainly: OFT is more fragile (frozen 0/20 on all three faults), recovers less on rotation (55 % vs 85 %) — its r_y fault is only 23 % identified where $\pi_{0.5}$ reached 90–100 % — and the healthy control lost one episode to the law itself, a sustained phantom that a confidence gate evaluated offline does not remove (§25.2 of the record).

Second suite, same calibration. The three headline cells replicate on `libero_object` without re-identifying W or M : rotation $0/20 \rightarrow 10/20$ ($p = 0.002$), translation $0/20 \rightarrow 15/20$ ($p = 6.1 \times 10^{-5}$), gain $0/20 \rightarrow 19/20$ ($p = 3.8 \times 10^{-6}$), zero regressions. Together with the backbone result: **one healthy-data calibration is a property of the robot**, whichever policy drives it and whatever it is asked to do.

A competence floor is not a repair result. Extending the headline to `libero_90` gave $4/20 \rightarrow 5/20$, $p = 1$. The unfaulted control on the same episodes is $6/20$: the checkpoint was fine-tuned on the four standard suites only and fails 14 of these 20 tasks with no fault present. There is nothing to restore. This is a fact about the policy, reported as such and not tabulated as a repair result — and the mirror of the ceiling artefact of §14: a null against a *floor* is not a null either. The healthy-frozen control on the exact episodes is now a prerequisite for every cell.

18 A second manipulator: ALOHA in joint space

Bimanual ALOHA (two ViperX arms) in `gym_aloha`, driven by `pi0_aloha_sim`. The action interface is **14 absolute joint targets in radians** at 50 Hz — not Cartesian, nothing rotating, so the $SO(3)$ machinery of §9 is absent by construction. The plant is re-identified here: a per-joint FIR on *position* rather than increment (a servo tracks an absolute target, so a target offset lives in the steady-state position error and leaves almost nothing in Δq), fitted on eight healthy rollouts, giving $R^2 = 0.989$ – 1.000 and $M \approx I$ with $\text{cond} = 7.3$.

Identification transfers. Separation against a matched healthy control, on the faulted joints:

fault	identification	leak onto clean joints
+0.02 rad on joints 0–5	94–106 %	≤ 0.007 rad
+0.05 rad on joints 0–5	84–91 %	≤ 0.007 rad
+0.10 rad on joints 0–5	69–71 %	—

Repair needs one more condition, and the task supplies it. Transfer-cube tolerates remarkably little: a static-correction sweep shows the frozen policy recovers nothing until the residual offset is under ≈ 0.5 cm at the gripper (-0.041 rad: $0/20$; -0.045 : $0/20$; exact -0.050 : $4/20$ against a healthy $5/20$). The policy itself succeeds only 25 % unfaulted. Against that margin the continuously updating law fails — and not because of its accuracy:

scheme	within-episode wander	residual	success
frozen, faulted	—	2.12 cm	0/20
adaptive, updating throughout	0.43 cm	0.17 cm	0/20
freeze after 30 / 100 / 200 steps	—	0.13–0.49 cm	0/20
adapt episode 0, then hold	0.000 cm	0.12 cm	5/20
oracle, exact fault from step 0	0	0	8/20
healthy, no fault	—	—	5/20

The same correction magnitude repairs when stationary and fails when moving. Holding the estimate after a single sacrificial episode recovers **5 of 19** at $\hat{f} = 0.0189$ rad, 94 % of the true fault — the healthy-frozen rate, from a policy scoring zero.

The precondition, stated as a limit. On a tight-margin task the estimate must be identified in a sacrificial episode and then frozen, not updated continuously. This was invisible on LIBERO, where a 20 Hz Cartesian reach tolerates centimetres of jitter and the law’s 0.019 cm/step of motion cost nothing. For a persistent hardware fault the price is one episode, which is what any calibration costs.

A criterion that predicts both outcomes. Both robots’ results follow from one comparison, measurable *before* any repair run and without knowing the fault: the task’s spatial margin (from a static-correction sweep) against the correction’s within-episode variation (from the estimator’s trajectory on one faulted episode). Repair works when the variation fits inside the margin.

	task margin	correction wander	outcome
LIBERO, translation 0.15	$\gg 1$ cm	0.019 cm/step	repair, 4/20 $\rightarrow$ 19/20
ALOHA, updating	0.5 cm	0.43 cm	no repair, 0/20
ALOHA, held	0.5 cm	0.000 cm	repair, 0/20 $\rightarrow$ 5/20

19 Negative results worth keeping

Naive error signal, no plant model. Using $\Delta x/\text{output_max} - a$ gives a nominal error of -0.175 on dx when it should be 0. Differencing faulted against nominal recovered ratios of 0.39, 2.46, -2.85 — noise. *Controller lag reads as a fault unless the plant is modelled.*

Robustness constants set without measuring. The bare law drifted — dz reached 0.642, **13 $\times$ truth**. The first fix made it *worse*: a deadzone of 0.05 against a typical residual norm of 0.034 suppressed nearly every update, ending several episodes at $\hat{f} = 0$ exactly. Rescaling to the measured residual fixed both; raising the normalisation constant ($0.05 \rightarrow 0.15$, cutting a systematic 32% attenuation to $\approx 5\%$) took the result from 70% to 87%.

A coupled MIMO plant: better fit, worse extrapolation. Wrist rotation is driven substantially by *translation* commands, which a per-axis model cannot represent. A coupled plant improves leave-one-episode-out held-out R^2 from $0.107 \rightarrow 0.615$ on dr_y . **And it made the closed loop worse:** 13/15 ($p = 0.109$) with dy estimated at -0.043 against a true $+0.050$ — a sign error the per-axis model never makes — versus 14/15 ($p = 0.014$). Forty-three parameters fitted on three nominal episodes extrapolate badly once the correction moves the executed command off that distribution, and *the adaptive law lives out of distribution by construction*. Kept as `--mimo`, default off.

Excitation is not the limiting factor for rotation. The policy essentially never commands rotation: translation command sd is 0.32–0.55 with 49–74% of steps above 0.15, while all three rotation axes sit at sd 0.02–0.055 with **0.0%** above 0.15. But excitation alone does not explain dr_y : it has the most rotation excitation and the worst fit, while dr_z has less and fits at 0.972. Coupling, not excitation, is the cause.

The recurring lesson. An improvement measured in one regime — open-loop fit, or excitation — can cost you in the regime that actually matters. The MIMO plant and the dither result have the same shape.

20 The video

results/phase05/adaptive_vs_frozen.mp4 shows two runs side by side: the same frozen model, the same task, the same injected offset fault. The only difference is whether the online repair is active. One run is the frozen policy — the fault has shifted its actions and it fails the task. The other is the same weights with the adaptive law running: it identifies the fault from the robot’s own motion within the first steps and cancels it, and the task completes normally. Nothing is retrained between the two runs.

A companion clip, adaptive_vs_frozen_offset010.mp4, shows the harder severity (offset 0.10), where frozen success is 0/8.

21 Limits and what this does not establish

- **Simulation.** Four LIBERO suites; hardware validation is in progress and has already produced one negative — the hardware plant is run-varying on the load-bearing joints, so the offline identification recipe will not port unchanged.
- **The law is blind to sensor bias.** $\pi_{0.5}$ ignores proprioception entirely, so a proprioceptive fault produces no change in the policy’s behaviour for the estimator to detect.
- **Gain faults are not settled.** One severity, $n=10$, no resolvable success difference; only the β estimates carry signal.
- **Rotation remains partly unidentified.** dr_y still carries the wrong sign in the gain law and fits at $R^2 = 0.336$.
- **M is identified open loop** and assumed valid in closed loop, where the policy compensates and the arm visits different states.
- **A gap to the oracle remains.** The oracle reaches 100%, the adaptive law 95% on spatial and 73% pooled.
- **The site screen does not separate sites** at the sample size run; action_out_proj/bias was chosen structurally.

22 Summary

1. A frozen 3.35 B VLA under a Cartesian action-interface fault drops from 99 % task success to 45 %, and at higher severities to *zero*.
2. The fault is **identified and cancelled online, within a single episode**, from the robot’s own motion — no gradients, no reward, no search over task success, and no learning across episodes: the estimate resets to zero every episode and reconverges in ~ 15 control steps.
3. **Four suites, paired, each individually significant:** pooled $28/120 = 23\% \rightarrow 78/120 = 65\%$, exact McNemar $p = 6.1 \times 10^{-13}$; 51 episodes fixed, one broken. The long-horizon suite goes from a frozen policy that never succeeds in forty episodes to fifteen recoveries.
4. **Every fault family tested is repaired:** constant additive (translation $4/20 \rightarrow 19/20$; rotation $0/20 \rightarrow 17/20$), structured mixed-sign, mid-episode onset, ramp, oscillatory, intermittent, and multiplicative loss of effectiveness ($0/20 \rightarrow 17/20$ at 80 % authority loss). Across **690 paired episodes: 275 fixed, 7 broken** (1.0 % regression) — and the three headline cells replicate on a **second backbone** (OpenVLA-OFT) and a **second suite** with the $\pi_{0.5}$ -identified calibration unchanged (§17).

5. **The identification machinery is load-bearing.** A tuned integral controller on the raw motion error, given five gains and reported at its best, reaches 35 % where this method reaches 95 %: the measured plant gain is 0.23, so raw-error integration chases a phantom $2.5\times$ the fault and opposite in sign.
6. **The boundary is the plant, not the estimator.** A fault spread across all six axes at high severity drives the arm into contacts where superposition fails; identically-corrected rotation faults recover 45 % alone and 10 % with an untouched translation component added.
7. **A second manipulator.** On 14-DOF bimanual ALOHA in joint space the law identifies a fault to 94 % from eight healthy rollouts and repairs transfer-cube to the healthy rate ($0/20 \rightarrow 5/20$) — but only when the correction is held stationary after one sacrificial episode. A continuously updating correction of the same magnitude repairs nothing, because its 0.43 cm of within-episode wander exceeds the task’s 0.5 cm margin. One measurable comparison — task margin against correction wander — predicts both robots’ outcomes before any repair run (§18).
8. **Two of this work’s own explanations were refuted by its own measurements** and are retained as such: that identifiability is predictable in advance from healthy action statistics, and that additive and multiplicative faults are identifiable on complementary channels. A third — that translation correction contributes nothing — proved to be an artefact of testing only at a severity the policy shrugs off.
9. Getting there required a metric bug on $SO(3)$ that had silently corrupted four results, a matched no-fault control that revealed half of a convincing estimate was bias, and three separate occasions where an estimate reading exactly its own projection bound was nearly reported as a measurement.